A Graph-Based Digital Twin of Conceptual Understanding

A concept is modeled as a graph of subconcepts (nodes) and relationships (edges). An AI tutor maintains an evolving estimate of the learner's understanding.

01 / THREE REPRESENTATIONS

E

REFERENCE KNOWLEDGE SPACE

$E = (V_E, R_E)$

Graph of target subconcepts and their relationships.

F_t

ACTUAL LEARNER SPACE

$F_t = (V_{F(t)}, R_{F(t)})$

Latent understanding that evolves through learning.

F'_t

ESTIMATED LEARNER SPACE

$F'_t \approx F_t$

Computational model inferred from observable evidence.

02 / LEARNING OBJECTIVE

$$\text{Minimize } D(E, F_t)$$

The distance function D remains to be defined. Since F_t cannot be measured directly, the AI uses F'_t to estimate knowledge gaps, select instruction, and assess changes.

03 / CONTINUOUS ADAPTATION

Reference E — target structure › **Teach** — AI action › **Observe** — learner evidence › **Update F′** — estimated state

Instruction changes the latent learner state; answers, explanations, and task performance supply evidence for revising the estimated state. The cycle then repeats.

04 / RESEARCH BASIS & OPEN QUESTION

Relevant traditions include concept mapping, learner modeling, knowledge tracing, and adaptive instruction. The accuracy and usefulness of the proposed graph-based twin must still be empirically validated.

Core question: How can F′ be constructed and continuously updated from learner behavior so that adaptive teaching reliably reduces the discrepancy between E and F(t)?

Scope: The digital twin is an uncertain estimate, not a readout of the mind. E is a chosen reference model.